Double Machine Learning with Interactive Fixed Effects

Working Paper

Binzhi Chen Annalivia Polselli Paul S. Clarke

Institute for Social and Economic Research, University of Essex

May 20, 2026

Motivation: Two Challenges

- Challenge 1: High-dimensional controls
  Modern applications involve **many control variables** $(p \geqslant NT)$ entering through **unknown, nonlinear** functions

- Challenge 2: Latent common factor shocks
  Panels share **unobserved time-varying factors**, global recessions, commodity cycles, pandemic effects — and units react heterogeneously.

This paper develops a framework that handles both.

Building Block 1: Double Machine Learning

Partially Linear Model — estimate θ_0, partial out controls X (Robinson, 1988)

$$Y_{it} = \theta_0 D_{it} + \underbrace{l_0(X_{it})}_{\text{nonlinear nuisance}} + \varepsilon_{it}, \qquad D_{it} = \underbrace{m_0(X_{it})}_{\text{nonlinear nuisance}} + V_{it}$$

DML procedure (Chernozhukov et al., 2018):

1. Use ML to estimate $\hat{l}(X)$ and $\hat{m}(X)$ on held-out data (cross-fitting)
2. Residualise: $\tilde{U} = Y - \hat{l}(X), \quad \tilde{V} = D - \hat{m}(X)$
3. Regress $\tilde{U}$ on $\tilde{V} \;\Rightarrow\; \hat{\theta}$

Key Result

Works with **any** sufficiently accurate ML learner. Cross-fitting removes overfitting bias. Valid inference under weak assumptions.

Building Block 2: Interactive Fixed Effects

Standard FE (additive): $\alpha_i + \gamma_t$

- Removes time-invariant differences
- Units react *identically* to time shocks

Interactive FE (Bai, 2009):

$$y_{it} = D_{it}'\theta + \lambda_i' f_t + \epsilon_{it}$$

$\lambda_i' f_t$ (multiplicative)

- f_t: latent common factors (recession, pandemic)
- λ_i: unit-specific loadings
- Nests additive FE; **cannot be removed by demeaning**

$\lambda_i' f_t$ captures this heterogeneous exposure to common shocks, confounds $\hat{\theta}$ if ignored.
Ignoring IFE biases the treatment effect: the latent factor variation is correlated with D_{it}

The Gap We Fill

	High-dim X & nonlinear	Interactive FE	Inference
OLS	$\times$	$\times$	$\checkmark$
IFE (Bai, 2009)	$\times$	$\checkmark$	$\checkmark$
DML-FE (Clarke and Polselli, 2025)	$\checkmark$	$\times$	$\checkmark$
CCE-Lasso (Rücker et al., 2025)	$\times$	$\checkmark$	$\checkmark$
DML-IFE (this paper)	$\checkmark$	$\checkmark$	$\checkmark$

This paper...

- Projection-based CCE factor removal + Neyman-orthogonal DML
- Any ML learner; p can exceed NT
- $\sqrt{NT}$-consistent; asymptotically normal

The Model: Partially Linear Panel with IFE

Outcome and treatment equations:

$$Y_{it} = \theta_0 \, D_{it} + \underbrace{l_0(\mathbf{X}_{it})}_{\text{nonlinear nuisance}} + \underbrace{\lambda_i' f_t}_{\text{interactive FE}} + U_{it} \tag{1}$$

$$D_{it} = m_0(\mathbf{X}_{it}) + \phi_i' f_t + V_{it} \tag{2}$$

Covariate factor structure: $\mathbf{X}_{it} = \Gamma_i' f_t + E_{it}$

Key variables:

- θ_0: treatment effect of interest
- $f_t \in \mathbb{R}^r$: latent common factors
- $\lambda_i, \phi_i, \Gamma_i$: unit-specific loadings
- p can exceed NT (high-dimensional)

Factor Removal: The CCE Projection

Based on (Pesaran, 2006; Rücker et al., 2025): cross-sectional averages $\bar{X}_t = N^{-1} \sum_i X_{it}$ approximately span the factor space:

$$\bar{X}_t \approx \bar{\Gamma}' f_t \quad \Rightarrow \quad \text{project onto } \operatorname{col}(\bar{X}) \text{ to remove } f_t$$

Projection matrix construction:

1. Threshold eigenvalues of $T^{-1}\bar{X}'\bar{X}$ to estimate $\hat{r}$
2. High-dim case $(p > T)$: extract top $\hat{r}$ eigenvectors $\hat{U}$; set $\hat{W} = \bar{X}\hat{U}$
3. Annihilator: $\hat{\Pi} = I_T - \hat{W}(\hat{W}'\hat{W})^{-1}\hat{W}'$

Advantage: $\hat{\Pi}F \approx 0$: projecting Y and D through $\hat{\Pi}$ removes the interactive FE. No non-convex optimisation; no iterative PCA.

The Panel DML-IFE Algorithm

- **Step 1: CCE Projection** Compute $\hat{\Pi}$ from cross-sect. avg. of X to remove $\lambda_i' f_t$
- **Step 2: Transform Data** $\tilde{Y}=\hat{\Pi}Y$, $\tilde{D}=\hat{\Pi}D$ (IFE-free data)
- **Step 3: Cross-fit DML** ML residuals on X Neyman score $\Rightarrow \hat{\theta}$

- **Block cross-fitting:** K folds, nuisance estimated on complement set
- **Any ML algorithm:** Lasso, gradient boosting, neural networks

Theoretical Guarantee (Wait for Theroem)

Monte Carlo: Design

Data-generating process:

- $r = 2$ latent factors; $\theta_0 = 1$;
- 100 replications
- Three DGP designs (see next slide):
  - DGP1: linear nuisance functions
  - DGP2: nonlinear, non-smooth
  - **DGP3: nonlinear, discontinuous**

Estimators:

- IFE (Bai, 2009) (benchmark)
- Panel DML-IFE (Lasso)
- Panel DML-IFE (XGBoost)
- Panel DML-IFE (NNet)

Dimensions:

N: $\{20,\ 100,\ 500\}$
T: $\{30,\ 50,\ 100\}$
p: $\{50,\ 100,\ 300\}$

Monte Carlo: DGP Specifications

Only $X_{it,1}$ and $X_{it,2}$ (of p covariates) enter the nuisance functions. Parameters: $a_1 = b_2 = 0.25$, $a_2 = b_1 = 0.5$.

DGP1:

$$l_0 = a_1 X_1 + a_2 X_2$$
$$m_0 = b_1 X_1 + b_2 X_2$$

DGP2:

$$l_0 = a_1 \max(X_1, 0) + a_2 |X_2|$$
$$m_0 = b_1 \mathbb{1}\{X_1 > 0\} + b_2 \log(1 + |X_2|)$$

DGP3:

$$l_0\left(\boldsymbol{X}_{it}\right) = a_1\left(X_{it,1} \cdot X_{it,2}\right) + a_2\left(X_{it,2} \cdot \mathbb{1}\{X_{it,2} > 0\}\right)$$
$$m_0\left(\boldsymbol{X}_{it}\right) = b_1\left(X_{it,1} \cdot \mathbb{1}\{X_{it,1} > 0\}\right) + b_2\left(X_{it,1} \cdot X_{it,2}\right)$$

MC and Empirical Need to Discuss with the Team

Results: DGP3, $N = 20$ (Small Panels)

Bias $= \hat{\theta} - 1$**; columns:** $p = \{50, 100, 300\} \times T = \{30, 100\}$**:**

	$p = 50$		$p = 100$		$p = 300$	
Estimator	$T = 30$	$T = 100$	$T = 30$	$T = 100$	$T = 30$	$T = 100$
IFE	0.245	0.207	0.293	0.215	0.295	0.243
DML-IFE (Lasso)	0.126	0.105	0.083	0.061	0.063	0.033
DML-IFE (XGBoost)	0.039	0.038	0.032	−0.027	0.044	−0.024
DML-IFE (NNet)	−0.151	0.005	−0.357	−0.192	−0.057	−0.627

- IFE: biased (≈ 0.20–0.30), worse at high p when N is small
- DML-IFE Lasso/XGBoost: bias shrinks as $T\uparrow$, $p\uparrow$ even with $N = 20$
- NNet: high variance at small N — large negative bias at high p

Results: DGP3, $N = 100$ (Medium Panels)

Bias $= \hat{\theta} - 1$**; columns:** $p = \{50, 100, 300\} \times T = \{30, 100\}$**:**

Estimator	$p = 50$, $T = 30$	$p = 50$, $T = 100$	$p = 100$, $T = 30$	$p = 100$, $T = 100$	$p = 300$, $T = 30$	$p = 300$, $T = 100$
IFE	0.202	0.201	0.200	0.199	0.202	0.201
DML-IFE (Lasso)	0.118	0.094	0.084	0.058	0.058	0.030
DML-IFE (XGBoost)	0.101	0.064	0.038	0.020	−0.003	0.000
DML-IFE (NNet)	0.071	0.086	−0.055	0.031	−0.459	−0.296

- IFE: **persistent bias** ≈ 0.20 regardless of T, p
- DML-IFE (Lasso/XGBoost): **bias shrinks** as $T\uparrow$ and $p\uparrow$ $\Rightarrow$ consistent
- DML-IFE (NNet): unstable at high p — large negative bias when $p = 300$

Results: DGP3, $N = 500$ (Large Panels)

Bias $= \hat{\theta} - 1$; **columns:** $p = \{50, 100, 300\} \times T = \{30, 100\}$:

	$p = 50$		$p = 100$		$p = 300$	
Estimator	$T = 30$	$T = 100$	$T = 30$	$T = 100$	$T = 30$	$T = 100$
IFE	0.198	0.199	0.199	0.199	0.198	0.199
DML-IFE (Lasso)	0.116	0.092	0.079	0.057	0.055	0.030
DML-IFE (XGBoost)	0.084	0.072	0.047	0.033	0.026	0.003
DML-IFE (NNet)	0.105	0.084	0.061	0.039	-0.107	-0.009

- IFE bias remains ≈ 0.20 even at $N = 500$ — N does not resolve IFE bias
- DML-IFE (Lasso/XGBoost): consistent improvement; XGBoost near-zero at high p
- NNet stabilises at large N — much less negative bias at $p = 300$ vs $N = 20$

Empirical Application: Asset Pricing

Research question:
Which firm characteristics have a *causal* effect on monthly excess stock returns?

Data:

- $N = 29$ Dow Jones large-cap stocks
- $T = 72$ months (April 2016 – March 2022)
- $p = 89$ lagged firm characteristics as controls

Treatment variables:

1. Firm size (market cap)
2. Bid-ask spread
3. Employee growth rate
4. Inventory change
5. Dividend omission
6. Earnings forecast
7. Turnover volatility

Estimators: OLS, IFE, DML-FE, DML-IFE (Lasso, Boosting, NNet)

Economic relevance: latent factors include global risk, liquidity cycles, macro shocks — conditions where IFE is crucial.

Empirical Results: Selected Firm Characteristics

Variable	OLS	IFE	DML-IFE (Lasso)	DML-IFE (NNet)
Panel A: Firm Size				
Coefficient	-0.008	-0.034^{**}	-0.007	-0.007
(Std. err.)	(0.008)	(0.016)	(0.004)	(0.006)
Panel B: Bid-Ask Spread				
Coefficient	$\mathbf{0.946}^{***}$	-0.207	$\mathbf{0.076}^{***}$	0.530^{***}
(Std. err.)	(0.222)	(0.343)	(0.025)	(0.193)
Panel C: Employee Growth Rate				
Coefficient	-0.009	-0.036	-0.004^{***}	-0.005^{***}
(Std. err.)	(0.025)	(0.023)	(0.000)	(0.002)

$N = 29$ firms, $T = 72$ months. Significance: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. SE: asymptotic (OLS); bootstrapped (IFE); cluster-robust at firm level (DML).

Empirical Results

Summary and Contributions

This paper...

1. **First DML framework for panels with Interactive Fixed Effects**
   Combines CCE-based projection with Neyman-orthogonal score construction

2. **Flexible and computationally tractable**
   Any ML learner; no iterative PCA; $\sqrt{NT}$-consistent

3. **Time and covariate dimensions drive consistency, not N**
   Enables reliable inference in macro-panels and rich-covariate settings

Thank You

Questions and Comments Welcome

Binzhi Chen
bc25911@essex.ac.uk

Annalivia Polselli
annalivia.polselli@essex.ac.uk

Paul S. Clarke
pclarke@essex.ac.uk

Institute for Social and Economic Research
University of Essex, Colchester, UK

References I

Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica*, 77(4):1229–1279.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters.

Clarke, P. S. and Polselli, A. (2025). Double machine learning for static panel models with fixed effects. *Econometrics Journal*, page utaf011.

Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4):967–1012.

Robinson, P. M. (1988). Root-n-consistent semiparametric regression. *Econometrica: journal of the Econometric Society*, pages 931–954.

Rücker, M., Vogt, M., Linton, O., and Walsh, C. (2025). Estimation and inference in high-dimensional panel data models with interactive fixed effects. *Quantitative Economics*, 16(4):1457–1509.